Bimlendra Ray

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Aspiring robotics engineer with hands-on experience in control systems development, modeling, and simulation using MATLAB/Simulink, ROS, MuJoCo and Gazebo. Skilled in model predictive control, state estimation, and inverse dynamics, proficiency in Python, MATLAB and C++. Passionate about designing adaptive biped robot architectures and optimizing simulator computation efficiency.

Education

University of Texas at Dallas *Richardson, TX, USA*

Master Of Science in Mechanical Engineering (DSC). GPA: 3.3 *May 2026*

Project: Congestion Control in V2X using DeePC

Coursework: Robust Control, Optimal Control, Optimization, Robot Control and Nav, Multi-agent system, Modeling and Simulation, MPC, Vehicle Dynamics

Uttarakhand Technical University *Dehradun, India*

Bachelor Of Technology in Mechanical Engineering. *2020*

Research work: Review paper on Composite material

NASA Higher Secondary School *Kathmandu, Nepal*

Coursework: Physics. *2016*

Research

Data-Enabled Predictive Control for Adaptive V2X Resource Allocation in High-Density Vehicular Networks *UT DALLAS*

Jan 2026 – Current

- Developing a **DeePC-based congestion control for V2V networks**, replacing RL with a data-driven predictive control using rate and transmit power as inputs and CBR/PDR/IPD as outputs.
- Tools: MATLAB, MPC, C++, Python

Projects

Hybrid Control of a Legged Robot in MuJoCo *Out of Curiosity*

Dec 2025 – Feb 2026

- Built a MuJoCo-based legged robot simulation with contact dynamics, actuator constraints, and phase-based hybrid control.
- Implemented a finite-state machine for stance, flight, touchdown, and liftoff transitions.
- Designed feedback controllers for height regulation, torso stabilization, and forward velocity tracking.
- Evaluated controller robustness under external push disturbances, mass variation, and ground-friction changes
- Generated plots for COM height, velocity tracking, joint motion, torque usage, and gait stability
- C++, MuJoCo, Hybrid Systems, Feedback Control, Simulation

End-to-End Autonomous Vehicle Control System (Quanser QCar) *UT DALLAS*
Dec 2025 – Feb 2026

- Designed and deployed a real-time autonomous vehicle control architecture integrating perception, planning, and control
- Implemented Model Predictive Control (ACADOS) for real-time trajectory tracking of vehicle lateral and longitudinal dynamics
- Integrated computer vision pipeline for traffic light and obstacle detection
- Optimized controller stability and compute latency for real-time execution under hardware constraints
- Developed full closed-loop autonomous navigation system operating in simulation.
- Tools: Python, ACADOS, OpenCV, MPC

Robust Stability and Synthesis of Self-Balancing Two-Wheeled Robot *UT DALLAS*
Feb 2026 – May 2026

- Developed a control-oriented uncertain state-space model of a two-wheeled self-balancing robot (TWSBR) with structured parametric uncertainty in payload mass and center-of-mass height using MATLAB
- Designed and compared LQR and mixed-sensitivity H_∞ controllers for robust stabilization, disturbance rejection, and actuator-constrained balancing control
- Performed robust stability and worst-case performance analysis using robstab, wcgain, Monte Carlo simulations, and worst-case uncertainty substitution to evaluate closed-loop behavior near instability boundaries
- Analyzed pole migration, sensitivity shaping, and uncertainty-induced performance degradation through time-domain and frequency-domain robust control methods
- Tools: MATLAB

Real-Time Neural Approximation of Constrained MPC via Imitation Learning *UT DALLAS*
Feb 2025 – May 2025

- Designed and implemented a constrained Model Predictive Controller (MPC) for an unstable dynamic system (Self-balancing two wheeled robots)
- Trained a deep neural network policy using behavior cloning to approximate the MPC feedback law
- Deployed the learned controller in closed-loop simulation and evaluated stability, tracking error, and disturbance rejection
- Analyzed robustness and distribution shift effects in closed-loop learned control
- Tools: PYTHON, MPC, Optimal Control, Robotics Simulation

Model Predictive Control for Disturbance-Aware Lap Time Optimization in Autonomous Racing Vehicles using two-step sequential programming *UT DALLAS*
Sep 2025 – Dec 2025

- Formulate the lap-time optimization problem under disturbances and model uncertainty.
- Develop an MPC that ensures the actual trajectory stays within a safety region around the nominal optimal racing line.
- Real time computation for obstacle avoidance, thanks to the ACADOS nonlinear optimization solver.
- Achieved computation time of 2ms which is fast enough for real time.
- Tools: MATLAB, SIMULINK, ACADOS

Optimal Trajectory following using AL-ILQR algorithms while avoiding dynamic obstacles using chance constraints — *UT DALLAS*
Feb 2025 – May 2025

- Developed an augmented AL-ILQR algorithm enabling optimal trajectory tracking while safely avoiding dynamic obstacles using chance-constraint formulations
- Modeled a linear quadrotor system, discretized the dynamics using RK4, and applied an LTV approximation to ensure real-time feasibility during constrained trajectory tracking.
- Demonstrated reliable obstacle avoidance performance in simulation; identified and analyzed causes of significant deviation from the reference trajectory to guide future controller refinement
- Tools: ILQR, Python

Detecting and Counting the number of objects on Production Line using PLC — *Wilkes University*
Feb 2024 – May 2024

- Developed PLC program and implemented on real hardware (Allen-Bradley)
- Detecting the number of objects in the container and automatically replacing it with empty using PLC controller Setting Alarming system to notify the task completion or any fault occurring during process

Skills & Interests

- Robotics & Autonomy:
    - Modeling and Control, Robust analysis and synthesis, Linear System analysis, State Estimation
    - MPC, Optimal Control, Data Driven Control
    - Autonomous Navigation, Trajectory Planning
- Programming:
    - Python, MATLAB, C++ (Beginner), SQL, PLC (Ladder logic)
- Robotics & Simulation:
    - MuJoCo, SIMULINK, ANSYS, ROS, GAZEBO, QUANSER QCAR
- AI/ML:
    - TensorFlow
- Design:
    - SolidWorks
- Hardware:
    - PLC (Allen-Bradley), Arduino

Experience

Silicon Soft and IT Consultant Pvt. Ltd. — *Kathmandu, Nepal*
Software Developer — *Jan 2022 – Dec 2023*

- Gain confidence in learning and doing the task assigned in time
- Improvised the Journey for Mobile Applications development for the first time in this company. Used Kotlin, Jetpack Compose and .Net framework.
- Worked on a Cross-platform Mobile application with other team members. Used SQL, Kotlin, KMM, KTOR
- Developed Government Web application using REACT and JS
- Handled two projects under my responsibilities and led the team member

Gorkha Brewery Pvt. Ltd. (Carlsberg Group) *Chitwan, Nepal*
Maintenance Engineer (Trainee) *Jul 2021– Dec 2021*

- Implemented Kaizen to find out the root problem in production line and presented a detailed solution
- Performed extended data acquisition and trend analysis to identify system-level energy losses and recommend efficiency improvements

Leadership & Awards

GORKHA BREWERY PVT. LTD. (PART OF CARLSBERG GROUP) *Chitwan, Nepal*
Kaizen star of the month November 2021 *Jul 2021 –Dec 2021*

- Modification of the pull of cap Shute to install empty warning alarm sensor to save beer.
- Used the Kaizen tool to find out the root cause of the frequent stoppage of the filler machine and modified the design. This idea increased production by 0.12 million additional bottles per year.